

Problem Descriptions

Line through two points. For $x_1 \neq x_2$,

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad b = y_1 - mx_1, \quad y = mx + b.$$

1. **Waiting for TransMilenio after class.** On weekday evenings, a university student leaves class and waits at a TransMilenio station near the Universidad Nacional. The wait always lasts between two and ten minutes, and every waiting time in that span is equally plausible. Determine the complete probability density rule for the waiting time, including what happens outside the stated span. Then find the chance that the student waits longer than seven minutes and the chance that the wait lasts from three through seven minutes.
2. **An evening commute across Bogotá.** A young apprentice travels by SITP from Kennedy to a training programme in Chapinero during the evening rush. Traffic can add anywhere from no delay to eight minutes to the usual journey. A delay of any length in that span is possible, and longer delays become steadily more common. An eight-minute delay is five times as common as no delay. Determine the complete probability density rule for the added travel time, including what happens outside the stated span. Then find the chance of more than six minutes of added time and the chance of between two and six minutes of added time.
3. **Weekly study at a public library.** Members of a Bogotá youth scholarship programme record the time they study each week at the Biblioteca Pública Virgilio Barco. Weekly totals range from two to ten hours. Totals become steadily more common from two hours up to a most common total of five hours, then steadily less common through ten hours; there are no totals right at either limit. Determine the complete probability density rule for weekly study time, including what happens outside the stated span. Then find the chance of studying for more than seven hours and the chance of studying from four through eight hours.
4. **A Sunday on the Ciclovía.** A group of young adults tracks how long its members spend cycling and meeting friends on Bogotá's Sunday Ciclovía. A recorded visit can last from zero to five hours. All durations up to two hours are equally common. All durations from two to five hours are also equally common, but they occur twice as often as durations in the shorter band. Determine the complete probability density rule for the visit length, including what happens outside the stated span. Then find the chance that a visit lasts more than four hours and the chance that it lasts from one through four hours.
5. **Time at a youth entrepreneurship fair.** Young adults attend a Bogotá entrepreneurship fair at Corferias for up to six hours. Immediate departures are not observed. Departures become steadily more common during the first two hours, remain equally common at their highest level from two through four hours, and then become steadily less common until six hours, when none are observed. Determine the complete probability density rule for the length of a visit, including what happens outside the stated span. Then find the chance of staying longer than five hours and the chance of staying from one through five hours.

Solutions

1. Uniform distribution

Support, shape, and normalization. On $[2, 10]$, the density is a rectangle of unknown height h . Total area 1 gives

$$1 = (10 - 2)h = 8h, \quad h = \frac{1}{8}.$$

Therefore the complete pdf is

$$f_T(t) = \begin{cases} \frac{1}{8}, & 2 \leq t \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

Tail probability. The right tail is a rectangle:

$$P(T > 7) = (10 - 7)\frac{1}{8} = \frac{3}{8} = 37.5\%.$$

Interval probability. The requested interval is also a rectangle:

$$P(3 \leq T \leq 7) = (7 - 3)\frac{1}{8} = \frac{1}{2}.$$

2. Linear distribution

Support, shape, and normalization. On $[0, 8]$, let the positive endpoint heights be h and $5h$. The entire density is a trapezoid, so

$$1 = \frac{1}{2}(8)(h + 5h) = 24h, \quad h = \frac{1}{24}, \quad 5h = \frac{5}{24}.$$

The line passes through $(0, 1/24)$ and $(8, 5/24)$. Its slope and intercept are

$$m = \frac{\frac{5}{24} - \frac{1}{24}}{8 - 0} = \frac{1}{48}, \quad b = \frac{1}{24}.$$

Hence

$$f_X(x) = \begin{cases} \frac{x+2}{48}, & 0 \leq x \leq 8, \\ 0, & \text{otherwise.} \end{cases}$$

Both endpoint heights are positive, so this nonconstant line is not triangular.

Tail probability. Since $f_X(6) = 1/6$ and $f_X(8) = 5/24$, the tail is a width-2 trapezoid:

$$P(X > 6) = \frac{1}{2}(8 - 6) \left(\frac{1}{6} + \frac{5}{24} \right) = \frac{3}{8}.$$

Interval probability. Here $f_X(2) = 1/12$ and $f_X(6) = 1/6$:

$$P(2 \leq X \leq 6) = \frac{1}{2}(6 - 2) \left(\frac{1}{12} + \frac{1}{6} \right) = \frac{1}{2}.$$

3. Triangular distribution

Support, shape, and normalization. The support is $[2, 10]$, the endpoint heights are 0, and the mode is 5. If the peak height is H , then

$$1 = \frac{1}{2}(10 - 2)H = 4H, \quad H = \frac{1}{4}.$$

The rising segment joins $(2, 0)$ to $(5, 1/4)$, and the falling segment joins $(5, 1/4)$ to $(10, 0)$. Their slopes are

$$m_1 = \frac{\frac{1}{4} - 0}{5 - 2} = \frac{1}{12}, \quad m_2 = \frac{0 - \frac{1}{4}}{10 - 5} = -\frac{1}{20}.$$

Using the zero-height endpoints gives the complete pdf

$$f_Q(q) = \begin{cases} \frac{q-2}{12}, & 2 \leq q \leq 5, \\ \frac{10-q}{20}, & 5 < q \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

The pieces meet at $f_Q(5) = 1/4$.

Tail probability. Since $f_Q(7) = 3/20$, the right tail is a triangle:

$$P(Q > 7) = \frac{1}{2}(10 - 7)\frac{3}{20} = \frac{9}{40} = 22.5\%.$$

Interval probability. At the outside boundaries, $f_Q(4) = 1/6$ and $f_Q(8) = 1/10$. Subtract the two tail triangles:

$$\begin{aligned} P(4 \leq Q \leq 8) &= 1 - \frac{1}{2}(4 - 2)\frac{1}{6} - \frac{1}{2}(10 - 8)\frac{1}{10} \\ &= \frac{11}{15} \approx 73.3\%. \end{aligned}$$

4. Piecewise-uniform distribution

Support, heights, and normalization. The support is $[0, 5]$, with breakpoint 2. The two constant heights are k and $2k$. Adding their rectangular areas gives

$$1 = 2k + (5 - 2)(2k) = 8k, \quad k = \frac{1}{8}.$$

Thus

$$f_W(w) = \begin{cases} \frac{1}{8}, & 0 \leq w < 2, \\ \frac{1}{4}, & 2 \leq w \leq 5, \\ 0, & \text{otherwise.} \end{cases}$$

Tail probability. The tail is a width-1, height-1/4 rectangle:

$$P(W > 4) = (5 - 4)\frac{1}{4} = \frac{1}{4}.$$

Interval probability. Split at the breakpoint and add two rectangles:

$$P(1 \leq W \leq 4) = (2 - 1)\frac{1}{8} + (4 - 2)\frac{1}{4} = \frac{1}{8} + \frac{1}{2} = \frac{5}{8}.$$

5. Piecewise-linear distribution

Support, heights, and normalization. The support is $[0, 6]$, with breakpoints 2 and 4 and heights 0, H , H , 0 at 0, 2, 4, 6. The total area is a triangle, a rectangle, and a triangle:

$$1 = \frac{1}{2}(2)H + (4 - 2)H + \frac{1}{2}(6 - 4)H = 4H, \quad H = \frac{1}{4}.$$

The first segment joins $(0, 0)$ to $(2, 1/4)$, so

$$m_1 = \frac{\frac{1}{4} - 0}{2 - 0} = \frac{1}{8}, \quad f_D(d) = \frac{d}{8}.$$

The middle segment joins $(2, 1/4)$ to $(4, 1/4)$ and is constant at $1/4$. The last segment joins $(4, 1/4)$ to $(6, 0)$, so

$$m_3 = \frac{0 - \frac{1}{4}}{6 - 4} = -\frac{1}{8}, \quad f_D(d) = \frac{6 - d}{8}.$$

Therefore the complete connected pdf is

$$f_D(d) = \begin{cases} \frac{d}{8}, & 0 \leq d \leq 2, \\ \frac{1}{4}, & 2 < d \leq 4, \\ \frac{6 - d}{8}, & 4 < d \leq 6, \\ 0, & \text{otherwise.} \end{cases}$$

The pieces meet at height $1/4$ at both breakpoints.

Tail probability. Since $f_D(5) = 1/8$, the right tail is a triangle:

$$P(D > 5) = \frac{1}{2}(6 - 5)\frac{1}{8} = \frac{1}{16} = 6.25\%.$$

Interval probability. The complement consists of two congruent tail triangles:

$$P(1 \leq D \leq 5) = 1 - \frac{1}{2}(1)\frac{1}{8} - \frac{1}{2}(1)\frac{1}{8} = \frac{7}{8} = 87.5\%.$$